



DPUSGEProjection

Projects 3D world coordinates to 2D screen coordinates for a given SGE view.

1. Inputs for each projection:

The names of these inputs start with *Coordi.* for the *i*-th coordinate set. It is possible to transform up to 10 coordinate sets (named *Coord1* up to *Coord10*) in parallel.

<i>X</i>	X coordinate of the object position in the 3D coord system
<i>Y</i>	Y coordinate of the object position in the 3D coord system
<i>Z</i>	Z coordinate of the object position in the 3D coord system

2. Static parameters:

<i>ViewID</i>	Number of the view for which the screen coordinates are computed. (Default: 0 = first view on the selected PC)
<i>OriginOnTop</i>	Specifies the location of the origin of the window coordinate system. It can be set to top left (<i>true</i>) or bottom left (<i>false</i>). Default: top left (<i>true</i>).
<i>NoClip</i>	Should coordinates be processed which cannot be seen in the given SGE view? (Default: false, which means that the output will be set to -9999 if the 3D coordinates are not visible.)

3. Outputs for each projection:

<i>WindowX</i>	Projected X coordinate of the object in window cords
<i>WindowY</i>	Projected Y coordinate of the object in window cords
<i>OffScreen</i>	Is set to true if the position is not visible in the SGE view, false if it is visible.

If the input position is currently not visible in this SGE view, the outputs depend on the setting of the *NoClip* parameter:

- If *NoClip* = false (the default), *WindowX* and *WindowY* are both set to -9999
- If *NoClip* = true, projected coordinates are available even if the position is not visible. However, they do not correspond to a window position, and may in some cases be completely invalid (e.g. if the input position is behind the viewer).

In both cases, the visibility of the input position can be checked using the *OffScreen* output.

4. Dependencies to other DPUs:

DPUSGEProjection must run together with DPUSGEView and DPUSGEWorld. The Index value of DPUSGEProjection must be larger than the Index values of DPUSGEView and DPUSGEWorld.



5. Functional principle:

You can test the DPU using the following script:

```
DPUSGEProjection Proj
{
    Computer = {VIS};
    Index = 12;
};

DPUSGEObject ProjObj
{
    Computer = {VIS};
    Index = 60;

    Object = "test.dot";

};

Connections = {
    Proj.Coord1.WindowX <-> ProjObj.X,
    Proj.Coord1.WindowY <-> ProjObj.Y
};
```

Instead of the object file "test.dot", an SGE object should be used which is positioned in the 2D overlay coordinate system. It will then appear at the screen location corresponding to the 3D world position Proj.Coord1.[XYZ].